

Michael Rizzo-Lopez

(previously Michael Rizzo Smith)

Curriculum Vitae

Nashville, TN — michael.rizzolopez@vanderbilt.edu — mjosmith.github.io

Computational astrophysicist specializing in hydrodynamic simulations of massive black hole binary dynamics in gaseous accretion disks and predicting their electromagnetic and gravitational-wave signatures for multi-messenger astrophysics.

EDUCATION

Ph.D Astrophysics Aug. 2023 - Exp. 2029
Vanderbilt University Nashville, TN
Department of Physics and Astronomy
Dissertation Title: *“Hydrodynamic Simulations of Relativistic Compact Object Inspirals in Accretion Disks”*
Advisor: *Kelly Holley-Bockelmann*

Bachelor of Science, Physics and Astronomy & Planetary Science Aug. 2017 - May 2021
Stony Brook University Stony Brook, NY
Department of Physics and Astronomy
Magna Cum Laude with Honors in Physics

PROFESSIONAL WORK AND AFFILIATIONS

NASA XRP 2026 Review Panel, Executive Secretary Mar. 2026
LISA Consortium, Core Member May 2025 - Present
Vanderbilt’s Graduate Student Astrophysics Journal Research Club, Jan. 2025 - Aug. 2026
Co-Coordinator
Rebuilt existing club structure to emphasize practical presentation skills and implemented a formal faculty feedback structure.
LISA Disc-IMRI Project Astrophysics Working Group, Code Captain Dec. 2023 - Dec. 2025
Establishing Multimessenger Astronomy Inclusive Training (EMIT), Sep. 2023 - Present
Graduate Student Scholar
ASAS-SN Project, Data Analyst Jun. 2021 - July 2023
Research Assistant for the All Sky Automated Survey for Supernovae at The Ohio State University.

AWARDS AND FELLOWSHIPS

Vanderbilt Physics and Astronomy Summer Research Award (\$2,000) May 2026
NASA CRESST II LISA Symposium Travel Award (\$1,500) Apr. 2026
NSF Graduate Research Fellow (\$150,000) Jun. 2025
Vanderbilt Graduate School Travel Grant (\$1,000) July 2024
McMinn Research Award (\$2,000) May 2024
Vanderbilt Robert T. Lagemann Award (\$3,500) Apr. 2024
Departmental Honors, Dept. of Physics May 2021
Stony Brook Campus Scholarship (\$26,000) Aug. 2017 - May 2021

PRESENTATIONS

Contributed Talk, The 16th International LISA Symposium June 2026
Presenting: *“Precessing, Eccentric Binary Black Hole Evolution in Accretion Disks”*
Contributed Talk, Vanderbilt’s Astrophysics Journal Research Club Nov. 2025
Presenting: *“Relativistic Binary Precession: Impact on Eccentric Massive Binary Black Hole Accretion and Hydrodynamics”*
Contributed Talk, Vanderbilt’s Astrophysics Journal Research Club Jan. 2025
Presenting: *“The dynamics and electromagnetic signatures of accretion in unequal mass binary black hole inspirals”*
Contributed Talk, The 15th International LISA Symposium July 2024
Presenting on behalf of the Disc-IMRI Code Comparison LISA Astrophysics Working Group

PUBLICATIONS

Reviewed Articles

(1 first-author, 5 co-author, 141 citations)

1. Andrea Derdzinski, Alexander J. Dittmann, Alessia Franchini, Alessandro Lupi, Noé Brucy, Pedro R. Capelo, Frédéric S. Masset, Raphaël Mignion-Risse, **Michael Rizzo Smith**, Edwin Santiago-Leandro, Martina Toscani, David A. Velasco-Romero, Robert Wissing, Mudit Garg, Lucio Mayer, Roberto Serafinelli, Lazaros Souvatzis, Daniel J. D’Orazio, and Jonathan Menu. “The LISA Astrophysics “Disc-IMRI” Code Comparison Project: Intermediate-Mass-Ratio Binaries in AGN-Like Discs”. In: *arXiv e-prints* (Dec. 2025), arXiv:2512.10893. arXiv: 2512.10893 [astro-ph.HE]
2. Kelly Holley-Bockelmann, Fazeel Mahmood Khan, Isaiah Williams, Jaelyn Roth, **Rizzo Smith, Michael**, Kaitlin Porter, Jillian Bellovary, Andrea Derdzinski, and Andrea V. Macciò. “Handy Relation between Binary Black Hole Merger Times and Host Galaxy Properties”. In: *Astrophysical Journal Letters* 995.1, L32 (Dec. 2025), p. L32. arXiv: 2508.14253 [astro-ph.GA]
3. J. M. M. Neustadt, C. S. Kochanek, and **Rizzo Smith, M.** “Constraints on pre-SN outbursts from the progenitor of SN 2023ixf using the large binocular telescope”. In: *MNRAS* 527.3 (Jan. 2024), pp. 5366–5373. arXiv: 2306.06162 [astro-ph.HE]
4. Matthew Kenworthy, [20 authors], and **Rizzo Smith, Michael**. “A planetary collision afterglow and transit of the resultant debris cloud”. In: *Nature* 622.7982 (Oct. 2023), pp. 251–254. arXiv: 2310.08360 [astro-ph.EP]
5. **Rizzo Smith, M.**, C. S. Kochanek, and J. M. M. Neustadt. “The late time optical evolution of twelve core-collapse supernovae: detection of normal stellar winds”. In: *MNRAS* 523.1 (July 2023), pp. 1474–1495. arXiv: 2212.09763 [astro-ph.HE]
6. A. Kawash, [11 authors], **Rizzo Smith, M.**, T. W. -S. Holoien, J. L. Prieto, and T. A. Thompson. “The Galactic Nova Rate: Estimates from the ASAS-SN and Gaia Surveys”. In: *The Astrophysical Journal* 937.2, 64 (Oct. 2022), p. 64. arXiv: 2206.14132

Non-Reviewed Articles

(6 first-author, 13 citations)

1. M. Rizzo Smith and ASASSN-Team. “An Update on ASASSN-21qj: A Rapidly Fading, Sun-Like Star; Back With a Vengeance”. In: *The Astronomer’s Telegram* 15531 (July 2022), p. 1
2. M. Rizzo Smith and ASASSN-Team. “ASASSN-22el: A Deep Eclipse Event”. In: *The Astronomer’s Telegram* 15308 (Apr. 2022), p. 1
3. M. Rizzo Smith and ASASSN-Team. “ASASSN-21sa: A Deep Dimming Event”. In: *The Astronomer’s Telegram* 14937 (Sept. 2021), p. 1
4. M. Rizzo Smith and ASASSN-Team. “ASASSN-21qj: A Rapidly Fading, Sun-Like Star”. In: *The Astronomer’s Telegram* 14879 (Aug. 2021), p. 1
5. M. Rizzo Smith and ASASSN-Team. “ASASSN-21nn: An Unusual Dimming Event in a Red Giant Star”. In: *The Astronomer’s Telegram* 14803 (July 2021), p. 1
6. M. Rizzo Smith and ASASSN-Team. “ASASSN-21ml: Discovery of an Extreme ($\Delta g > 10$ mag) L-Dwarf Flare”. In: *The Astronomer’s Telegram* 14778 (July 2021), p. 1

OUTREACH

Nashville Adventure Science Center, Way Late Play Date , Invited Speaker Public fundraising event dedicated to making science learning accessible for all. “Seeing” Black Holes (50+ attendees)	Apr. 2026
Astronomy on Tap Nashville , Host Facilitate event, run trivia, and answer astronomy related questions from the public. (100+ monthly attendees)	2025 - Present
10-Years of Gravitational Waves, Vanderbilt Community Celebration , Poster “It’s OK to Be Gassy: How Cosmic Gas Guides Black Hole Collisions” (50+ attendees)	Sep. 2025
NASA Advocacy Day , Vanderbilt Federal Relations Met with congressional staffers in Washington D.C. to emphasize funding public science and NASA.	July 2025
NSF GRFP: Navigating the Application Panel , Panelist Panel to prospective applicants (30+ attendees).	Oct. 2025
Vanderbilt Society of Physics Students , Invited Panelist Panel on graduate school application process (15 students).	Oct. 2025
EMIT Outreach , Invited Speaker Career talk on computational astrophysics and higher ed (50+ students).	Jan. 2025
Vanderbilt Society of Physics Students , Invited Panelist Panel on graduate school application process (20+ students).	Oct. 2024
Valor Collegiate Academy , Invited Speaker Career talk to first-year class (150+ students).	Mar. 2024

TEACHING

Graduate Teaching Assistant

Fall 2023 - Spring 2025

Course Title: ASTR 1010L: Intro to Astronomy Lab

Responsibilities: Course instructor for weekly 3.5 hour class, lectured, facilitated group work, and graded lab packets.

Number of Students: ~21 per term, 4 terms – Rated 5/5 Effective Instructor by 100% of Evaluation

Respondents (~ 50% of students taught).

Private Tutor

Feb. 2020 - Oct. 2020

Intro Physics for Medical Professionals

Undergraduate Teaching Assistant

Spring 2020

Course Title: AST 203: Astronomy